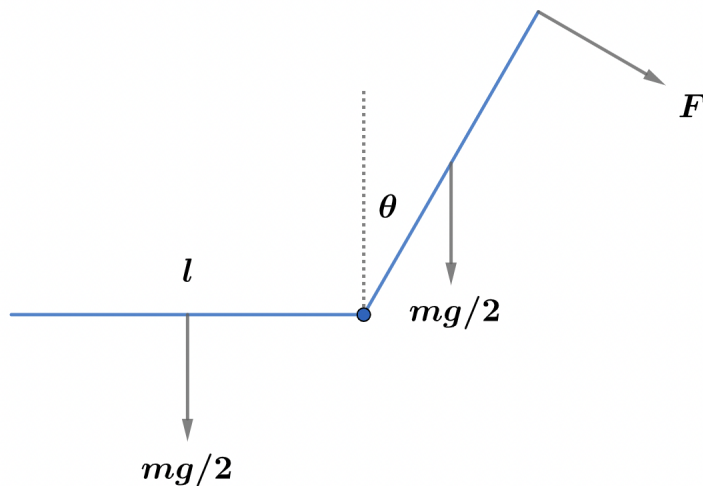


2022A F=ma Exam: Problem 13

Kevin S. Huang



When the bottom of the laptop is about to lift, the normal force only acts at the hinge. Taking the hinge as our pivot point, we have

$$\tau_{\text{net}} = -\frac{mg}{2} \left(\frac{l}{2} \right) + Fl + \frac{mg}{2} \left(\frac{l}{2} \right) \sin \theta = 0$$

$$Fl = \frac{mg}{2} \left(\frac{l}{2} \right) (1 - \sin \theta)$$

$$F = \frac{mg(1 - \sin \theta)}{4}$$

so the answer is C.